



Untitled31.ipynb ☆



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RAM

Disk



[5]

✓ 10m



```
image_path = None

for root, dirs, files_list in os.walk("blood_dataset"):
    for file in files_list:
        if file.lower().endswith((".jpg", ".jpeg", ".png")):
            image_path = os.path.join(root, file)
            break
    if image_path:
        break

image = cv2.imread(image_path)

image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)

gray = cv2.resize(gray, (256, 256))

_, segmented = cv2.threshold(gray, 120, 255, cv2.THRESH_BINARY)

input_layer = Input(shape=(256,256,1))

x = Conv2D(16,(3,3),activation="relu",padding="same")(input_layer)
x = MaxPooling2D((2,2))(x)
x = Conv2D(32,(3,3),activation="relu",padding="same")(x)
x = MaxPooling2D((2,2))(x)
x = Conv2D(64,(3,3),activation="relu",padding="same")(x)
x = UpSampling2D((2,2))(x)
x = Conv2D(32,(3,3),activation="relu",padding="same")(x)
x = UpSampling2D((2,2))(x)
output = Conv2D(1,(1,1),activation="sigmoid")(x)
```



Variables

Terminal



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Python 3

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model = Model(inputs=input_layer, outputs=output)

model.summary()

plt.figure(figsize=(10,4))

plt.subplot(1,2,1)

plt.imshow(gray,cmap="gray")

plt.title("Preprocessed Image")

plt.axis("off")

plt.subplot(1,2,2)

plt.imshow(segmented,cmap="gray")

plt.title("Segmented Output")

plt.axis("off")

plt.tight_layout()

plt.show()

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...

Input Size = 256 x 256

After 2 Pooling Layers = 64 x 64

After 2 Upsampling Layers = 256 x 256

Choose Files

archive (1).zip

archive (1).zip(application/x-zip-compressed) - 113386997 bytes, last modified: 7/3/2026 - 100% done

Saving archive (1).zip to archive (1).zip

Model: "functional_1"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 256, 256, 1)	0
conv2d_1 (Conv2D)	(None, 256, 256, 16)	160
max_pooling2d_1 (MaxPooling2D)	(None, 128, 128, 16)	0



```
Input Size = 256 x 256
After 2 Pooling Layers = 64 x 64
After 2 Upsampling Layers = 256 x 256
Choose Files archive(1).zip
archive (1).zip(application/x-zip-compressed) - 113386997 bytes, last modified: 7/3/2026 - 100% done
Saving archive (1).zip to archive (1).zip
Model: "functional_1"
```

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 256, 256, 1)	0
conv2d_1 (Conv2D)	(None, 256, 256, 16)	160
max_pooling2d_1 (MaxPooling2D)	(None, 128, 128, 16)	0
conv2d_2 (Conv2D)	(None, 128, 128, 32)	4,640
max_pooling2d_2 (MaxPooling2D)	(None, 64, 64, 32)	0
conv2d_3 (Conv2D)	(None, 64, 64, 64)	18,496
up_sampling2d (UpSampling2D)	(None, 128, 128, 64)	0
conv2d_4 (Conv2D)	(None, 128, 128, 32)	18,464
up_sampling2d_1 (UpSampling2D)	(None, 256, 256, 32)	0
conv2d_5 (Conv2D)	(None, 256, 256, 1)	33

Total params: 41,793 (163.25 KB)
Trainable params: 41,793 (163.25 KB)
Non-trainable params: 0 (0.00 B)

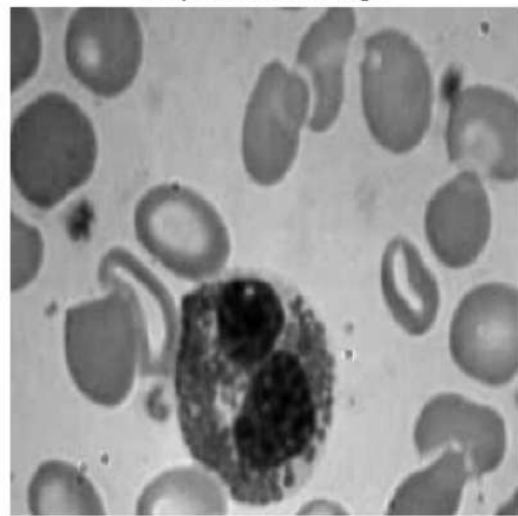




conv2d_3 (Conv2D)	(None, 64, 64, 64)	18,496
up_sampling2d (UpSampling2D)	(None, 128, 128, 64)	0
conv2d_4 (Conv2D)	(None, 128, 128, 32)	18,464
up_sampling2d_1 (UpSampling2D)	(None, 256, 256, 32)	0
conv2d_5 (Conv2D)	(None, 256, 256, 1)	33

Total params: 41,793 (163.25 KB)
Trainable params: 41,793 (163.25 KB)
Non-trainable params: 0 (0.00 B)

Preprocessed Image



Segmented Output

